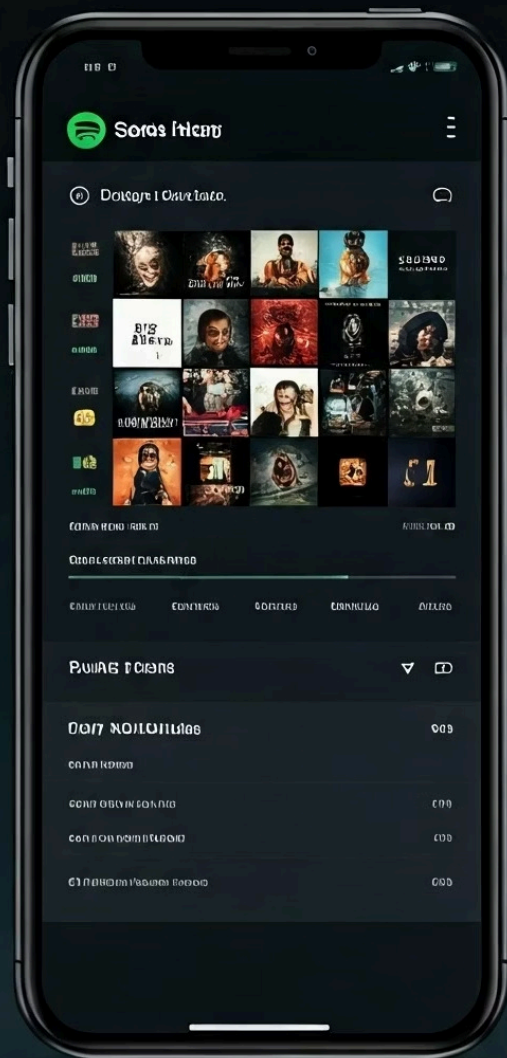




Turning Data Into Vibes: Building Spotify Playlists with KMeans Clustering

by Amruta Chaudhari, Katharina Gerhardt & Puyan Golestani

Our Approach



Data Scaling



Cluster Estimation (k) with Inertia, Elbow and Silhouette Method



Business Implications



Data Exploration and Evaluation

Spotify Song Features as the Basis for Clustering

Excluded: Key

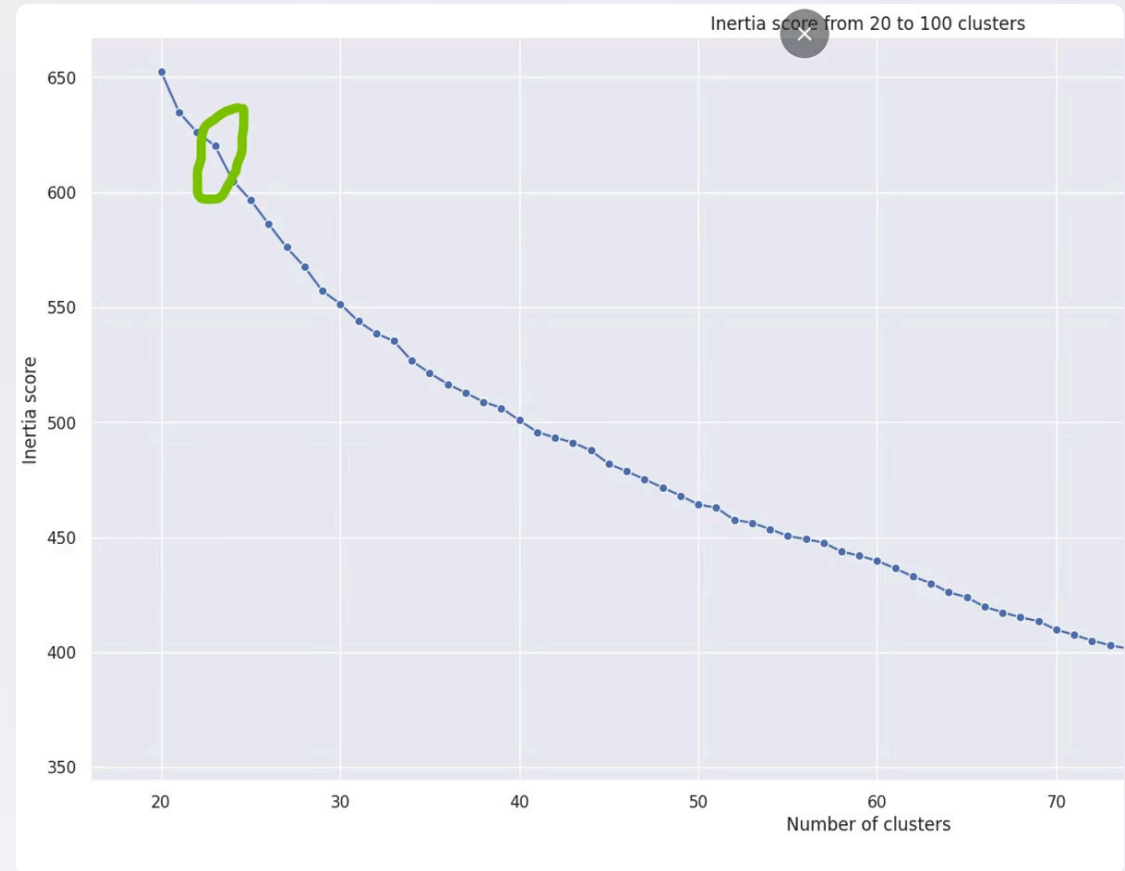


Feature Scaling

- MinMaxScale
- Audio features have originally different measuring
- Scaling to prevent dominance of certain features

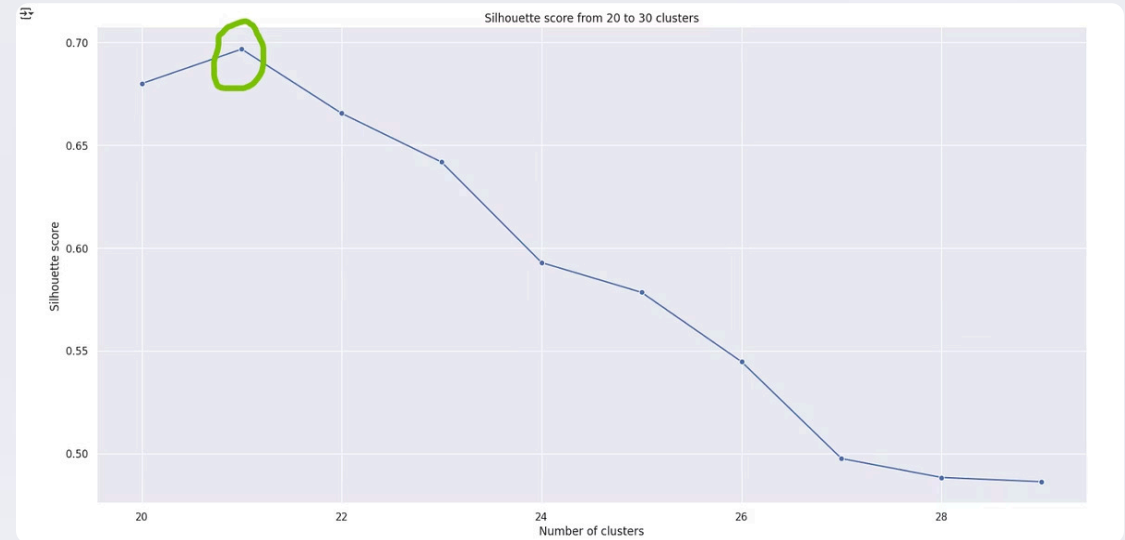
Inertia and Elbow Method

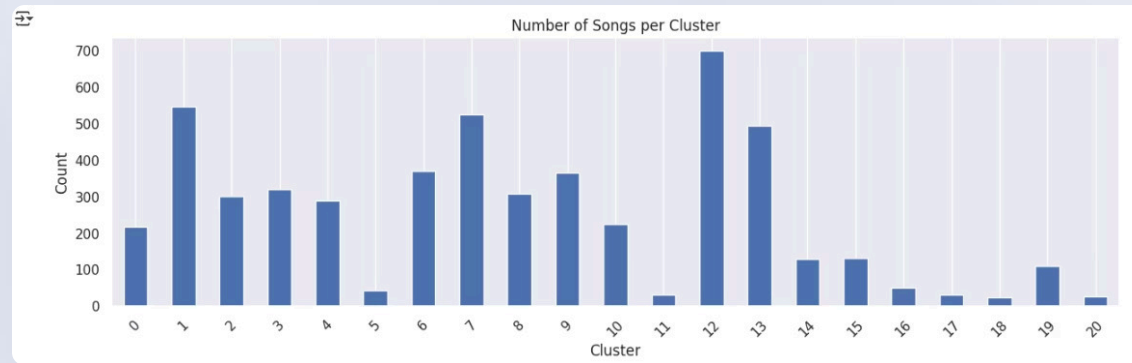
- Inertia measures distance from cluster point to cluster centre
- Elbow Method shows point where the rate of inertia sharply decreases



Silhouette Method

- measures how well data point fits into cluster compared to others
- best number of clusters (k) has highest average silhouette rate





Application of 21 Clusters

- too many songs in playlists
- lack of focus
- lower engagement

Business Considerations



Authenticity

playlists with distinguished moods and styles



A/B Testing

more granular testing possibilities



Targetted recommendations and placement

tracked easier in smaller playlists



Flexibility for Future Growth

merge, divide or adjust playlists

Öffentliche Playlist

After Work Vibes

Katharina Pralina Gepard • 7 Songs, 43 Min. 19 Sek.

Benutzerdefinierte Reihenfolge

#	Titel	Album	Hinzugefügt am	
1	 País Tropical Sergio Mendes & Brasil '77	País Tropical	vor 4 Minuten	2:49
2	 Chuck Berry Fields Forev... Gilberto Gil	Gilberto Gil (Ao Vivo em...	vor 4 Minuten	9:20
3	 Kenha Ki Ben Ki Ta Bai - L... Mayra Andrade	Studio 105 (Live)	vor 3 Minuten	5:30
4	 Aquele Abraço Gilberto Gil	Gilberto Gil (1969)	vor 3 Minuten	5:24
5	 Let There Be Peace On E... Ricky Dillard	Choirmaster	vor 2 Minuten	7:33
6	 More Abundantly Medle... Ricky Dillard	Let There Be Peace On E...	vor 2 Minuten	6:28
7	 Confidence - Live Tasha Cobbs Leonard	Grace (Live/Deluxe)	vor 47 Sekunden	6:15

Playlist Examples

- playlist high in genre mix (Gospel, Bossa Nova, Jazz)
- Brazilian artists could form different playlist (top 4)
- gospel songs might not be considered relaxing to some listeners

Conclusion

- Hard assignments only
- must choose k in advance
- limited use for music clustering

